COMP233 - Discrete Mathematics

Lecture 2.2
Conditional Statements

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The Logic of Compound Statements - Section 2.2

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ORIENTATION

Learning outcomes and lecture map

By the end of this lecture, you should be able to translate, evaluate, compare, and apply conditional statements.

Learning outcomes

- Evaluate p -> q and explain its single false case.
- Negate a conditional correctly and rewrite it as a disjunction.
- Construct the contrapositive, converse, and inverse.
- Interpret "only if", biconditionals, and necessary/sufficient conditions.
- Apply conditional logic to mathematical and computing examples.

Concept flow

1. **Meaning + truth table**
2. **Equivalent forms + negation**
3. **Contrapositive / converse / inverse**
4. **Biconditional + necessary/sufficient**
5. **Applications + concept checks**

MOTIVATION

A conditional is a rule, not automatically a cause

Start with familiar "if-then" statements, then separate everyday meaning from formal logic.

Example A

If 9 is divisible by 6, then 9 is divisible by 3.

hypothesis: 9 is divisible by 6

conclusion: 9 is divisible by 3

Example B - hypothetical promise

If you study, then you pass.

Treat this as a stated rule for logic practice - not a universal causal fact.

The formal connective p -> q answers one question:

When does the rule fail?

It fails exactly when p happens but q does not.

Formal implication is logical; it does not by itself assert causation.

CORE CONCEPT

Conditional statement: definition and truth table

The connective -> creates a new statement from a hypothesis and a conclusion.

Definition

If p and q are statements, the conditional **p -> q** is read "if p, then q."

p = hypothesis (antecedent)
q = conclusion (consequent)

Key idea

The conditional forbids exactly one situation: p is true while q is false.

p	q	p -> q
T	T	T
T	F	**F**
F	T	T
F	F	T

False iff p = T and q = F.

False hypothesis and vacuous truth

When the hypothesis is false, material implication is true regardless of the conclusion.

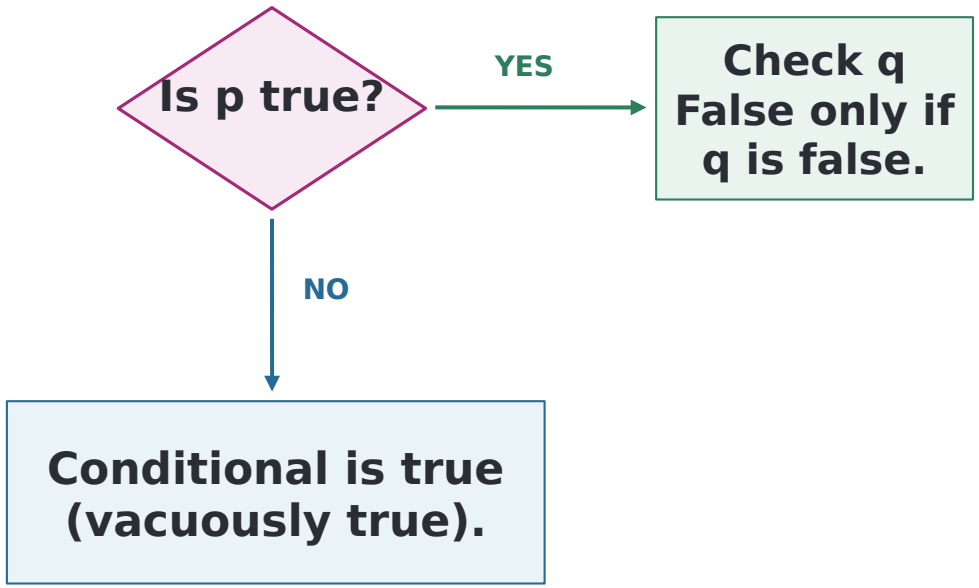
F -> T = T and F -> F = T

Source example

If 0 = 1, then 1 = 2.

The hypothesis 0 = 1 is false, so the conditional is true under the formal truth table.

Why? A conditional makes a commitment only when its hypothesis is true.



PROCEDURE

How to evaluate a compound conditional

Work from the inside out, then evaluate the main conditional last.

1
Identify the main connective.

2
Evaluate negations.

3
Evaluate AND / OR parts.

4
Evaluate -> using the single false case.

(p ∨ ¬q) -> ¬p

Decompose the statement before filling the final column:

¬p	¬q	p ∨ ¬q	(p ∨ ¬q) -> ¬p

Exam habit: mark the hypothesis and conclusion of the final -> before deciding T/F.

WORKED EXAMPLE

Worked truth table: (p ∨ ¬q) -> ¬p

The final implication is false exactly in rows where its hypothesis is true and its conclusion is false.

p	q	¬p	¬q	p ∨ ¬q	(p ∨ ¬q) -> ¬p
T	T	F	F	T	**F**
T	F	F	T	T	**F**
F	T	T	F	F	T
F	F	T	T	T	T

Rows 1-2: hypothesis = T and conclusion = F -> result F.

Rows 3-4: no violation -> result T.

Equivalent form: implication as a disjunction

A conditional can be rewritten without the -> connective.

$$p \rightarrow q \equiv \neg p \vee q$$

Interpretation
Either p is false, or q is true.

That is exactly the condition under which p -> q is true.

Example
"If the file is encrypted, then access is controlled."

Equivalent form: **"The file is not encrypted, or access is controlled."**

This equivalence is useful in Boolean algebra, SAT, and logical simplification.

Negating a conditional

To make "if p then q" false, p must occur and q must fail.

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

Derivation

$$\begin{aligned} & \neg(p \rightarrow q) \\ \equiv\ & \neg(\neg p \vee q) \\ \equiv\ & \neg(\neg p) \wedge \neg q \\ \equiv\ & p \wedge \neg q \end{aligned}$$

Do not negate it as $\neg p \rightarrow \neg q$.

That is the inverse, not the negation.

Negation practice: preserve the original examples

A correct negation asserts the hypothesis and denies the conclusion.

Original course example 1

If my lecture is at Masri 109, then I cannot buy coffee.

Negation: My lecture is at Masri 109 and I can buy coffee.

Original course example 2

If Amjad loves Zatar, then Amjad is smart.

Negation: Amjad loves Zatar and Amjad is not smart.

Concept check

Negate: "If the server authenticates the user, then access is granted."

Answer: **The server authenticates the user AND access is not granted.**

Division into cases

One implication from a disjunction can be split into two conditional cases.

$$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

Meaning
If either p or q is enough to reach r, then:

- p alone must imply r, and
- q alone must imply r.

Example
Let r = "the system enters safe mode."

If either overheating (p) or low battery (q) triggers safe mode, then each trigger separately must imply safe mode.

The contrapositive

Negate both parts and reverse their order.

Original: p -> q → **Contrapositive: ¬q -> ¬p**

Logical equivalence: p -> q ≡ ¬q -> ¬p

p	q	p -> q	¬q	¬p	¬q -> ¬p
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Contrapositive examples

A useful habit: name p and q first, then write ¬q -> ¬p.

Source example 1

If my lecture is at Masri 109, then I cannot buy coffee.

Contrapositive: If I can buy coffee, then my lecture is not at Masri 109.

Source example 2

If Amjad loves Zatar, then Amjad is smart.

Contrapositive: If Amjad is not smart, then Amjad does not love Zatar.

Mathematical example

If an integer n is divisible by 4, then n is even.

Contrapositive: If n is not even, then n is not divisible by 4.

Converse and inverse

Two related statements are easy to form - and easy to confuse with the original.

Original	Converse	Inverse
p -> q	q -> p	¬p -> ¬q

How to construct them

Converse: swap p and q.

Inverse: negate p and q without swapping them.

Contrapositive: swap AND negate.

Source example

If my lecture is at Masri 109, then I cannot buy coffee.

Converse: If I cannot buy coffee, then my lecture is at Masri 109.

Inverse: If my lecture is not at Masri 109, then I can buy coffee.

COMPARISON

Which related statements are equivalent?

The equivalence pattern is structural and worth memorizing.

Statement	Form	Relationship
Original	p -> q	≡ contrapositive
Contrapositive	¬q -> ¬p	≡ original
Converse	q -> p	≡ inverse
Inverse	¬p -> ¬q	≡ converse

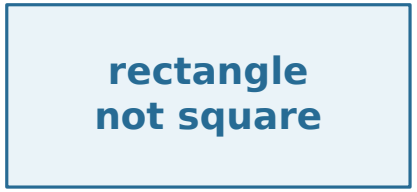
In general: original is NOT equivalent to converse or inverse.

Counterexample

Original: If a figure is a square, then it is a rectangle. TRUE

Converse: If a figure is a rectangle, then it is a square. FALSE

A non-square rectangle disproves the converse.



Language trap: "p only if q"

"Only if" points from the restricted statement to its requirement.

p only if q means p -> q (equivalently ¬q -> ¬p)

Interpretation

q is required for p.
If q does not occur, p cannot occur.

Caution: "p only if q" does NOT mean q -> p.

Course example

"A student may take COMP233 only if the prerequisite is satisfied."

Take COMP233 -> prerequisite satisfied

CORE CONCEPT

Biconditional: "if and only if"

A biconditional is true when its two component statements have the same truth value.

Definition

p ↔ q is read "p if and only if q" ("p iff q").

It asserts both directions.

Same truth value -> TRUE
Opposite truth values -> FALSE

p	q	p ↔ q
T	T	T
T	F	F
F	T	F
F	F	T

A biconditional is two conditionals

To prove or interpret p ↔ q, check both directions.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

Original course example: program correctness

"This computer program is correct if and only if it produces correct answers for all possible sets of input data."

Direction 1: If the program is correct, then it produces correct answers for all possible inputs.

Direction 2: If it produces correct answers for all possible inputs, then it is correct.

Necessary and sufficient conditions

These phrases are arrow-direction statements in disguise.

A is sufficient for B
A -> B
A guarantees B.

A is necessary for B
B -> A
Equivalently, ¬A -> ¬B.

A is necessary and sufficient for B
A ↔ B
Both A -> B and B -> A hold.

Shortcut: sufficient goes FORWARD; necessary points BACK from B to A.

English-to-symbol translation map

Translate the language first; then reason with the arrows.

English phrase	Symbolic form	What it means
If A, then B	A -> B	A is sufficient for B
A only if B	A -> B	B is necessary for A
A is sufficient for B	A -> B	A guarantees B
A is necessary for B	B -> A	B cannot occur without A
A iff B	A ↔ B	both directions
A necessary and sufficient for B	A ↔ B	A -> B and B -> A

Necessary and sufficient: examples and cautions

Use mathematical examples when you want the truth of the implication to be independent of a local policy.

Example 1 - sufficient

If n is divisible by 4, then n is even.

Divisible by 4 is sufficient for evenness.

Example 2 - necessary

If n is divisible by 4, then n is even.

Being even is necessary for divisibility by 4.

Source examples about passing a course or eligibility for an ID card are policy-dependent.

Treat them as stipulated rules only when the policy is explicitly given; they are not logical truths by themselves.

NOTATION

Order of operations for logical connectives

Use parentheses whenever an expression could reasonably be read in more than one way.

1	¬	**negation first**
2	∧ ∨	**AND / OR next**
3	-> ↔	**conditional / biconditional last**

Example: **¬p ∨ q ∧ r** is read as **(¬p) ∨ (q ∧ r)**. When -> and ↔ both appear, add parentheses to state the intended grouping.

Application: access-control policies

Necessary/sufficient language matters in cybersecurity because reversing an arrow changes the policy.

Policy statement

"Privileged access requires MFA."

Let A = privileged access
Let M = MFA completed

A -> M

MFA is NECESSARY for privileged access.

Wrong reversal
M -> A would say MFA alone guarantees privileged access.

Correct interpretation
Access may require MFA plus other conditions.

Application: conditionals in code and SAT

The same equivalences become practical transformations in program reasoning and Boolean solving.

Rule form

p -> q

Equivalent clause: **¬p ∨ q**

Failure condition

¬(p -> q)

Equivalent counterexample state: **p ∧ ¬q**

Why this matters

- SAT / Boolean simplification can replace implications with clauses.
- Software testing looks for states that violate an intended implication: p true while q false.

Synthesis: one statement, many equivalent views

Use this map to move between notation, English, and related statements.

p -> q

- **¬p ∨ q**
  implication as OR
- **¬q -> ¬p**
  contrapositive
- **¬(p -> q) ≡ p ∧ ¬q**
  negation
- **p only if q**

- **q -> p (converse)**
  NOT generally equivalent to original
- **¬p -> ¬q (inverse)**
  Equivalent to the converse

Next lecture: use these forms to test valid arguments and inference rules.

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Learning Appendix

Core concepts, connections, worked examples, knowledge check, and answer key.

Appendix: retrieval, transfer, and exam preparation

Appendix A - Core concepts and common misconceptions

What you should be able to recall without rebuilding the entire lecture.

Eight essentials

- p -> q is false only when p is true and q is false.
- False hypothesis -> the conditional is true.
- p -> q ≡ ¬p ∨ q.
- ¬(p -> q) ≡ p ∧ ¬q.
- p -> q ≡ ¬q -> ¬p.
- Converse and inverse are equivalent to each other, not generally to the original.
- p ↔ q means both directions.
- Sufficient = forward arrow; necessary = reverse requirement.

Misconception watch-list

- Treating p -> q as a causal claim.
- Thinking a false hypothesis makes the conditional false.
- Negating as ¬p -> ¬q instead of p ∧ ¬q.
- Assuming a true conditional has a true converse.
- Reading "only if" backwards.
- Confusing necessary with sufficient.
- Using iff when only one direction is known.

APPENDIX

Appendix B - Conceptual connections

A compact comparison matrix for the conditional family.

Name	Form	Equivalent to original?	How to build
Original	p -> q	Yes	given
Contrapositive	¬q -> ¬p	Yes	swap + negate
Converse	q -> p	No, in general	swap
Inverse	¬p -> ¬q	No, in general	negate
Biconditional	p ↔ q	Stronger: both directions	(p -> q) ∧ (q -> p)

Language bridge: "only if" -> implication; "necessary and sufficient" -> biconditional.

Appendix C - Additional worked examples

Three distinct cases: truth evaluation, language translation, and counterexample.

1. Truth value

p = F, q = F. Evaluate p -> q.

Reason: the hypothesis is false.

Result: TRUE.

2. Translate language

"Logging in is necessary for viewing grades."

Let V = view grades, L = log in.

Necessary means: V -> L

3. Disprove a converse

Original: divisible by 4 -> even.
Converse: even -> divisible by 4.

Counterexample: 6 is even but not divisible by 4.

So the converse is false.

APPENDIX

Appendix D - Knowledge check (1 of 2)

Answer before looking at the key. Explain your reasoning, not only the final symbol.

Q1. When is p -> q false?

Conditional truth | Easy | Understand

A) p=T,q=T B) p=T,q=F C) p=F,q=T D) p=F,q=F

Q2. "Access is granted only if authentication succeeds." Which form is correct?

Only-if translation | Medium | Apply

A) Auth -> Access B) Access -> Auth C) ¬Access -> ¬Auth D) Access ↔ Auth

Q3. True or false: A conditional is logically equivalent to its inverse. Justify.

Related statements | Medium | Analyze

Q4. Write the contrapositive: "If n is divisible by 6, then n is divisible by 3."

Contrapositive | Easy | Apply

Appendix D - Knowledge check (2 of 2)

Use counterexamples and equivalences when they are more efficient than a full truth table.

Q5. Error detection: "If n is divisible by 4, then n is even. n is even, so n is divisible by 4." Identify the error.

Converse error | Medium | Analyze

Q6. Give a counterexample to: "If an integer is even, then it is divisible by 4."

Counterexample | Medium | Create

Q7. Prove or disprove: "n is divisible by 4 iff n is even" for integers n.

Biconditional | Hard | Evaluate

Q8. A system policy says "Privileged access requires MFA." State which condition is necessary and which implication represents the policy.

Application | Medium | Apply

Appendix E - Answer key (1 of 2)

Check both the answer and the reasoning.

A1. B

p -> q is false only when p = T and q = F.
The other three rows are true.

A2. B

"Access only if authentication" means Access -> Auth.
A reverses the meaning; C is the inverse; D adds an unsupported reverse implication.

A3. False.

The inverse ¬p -> ¬q is not generally equivalent to p -> q. It is equivalent to the converse q -> p.

A4.

If n is not divisible by 3, then n is not divisible by 6.
This is ¬q -> ¬p.

Appendix E - Answer key (2 of 2)

Counterexamples are enough to disprove a universal converse or biconditional.

A5. Converse error

The argument replaces p -> q with q -> p. A true conditional does not guarantee its converse.

A6. Example: 6

6 is even but is not divisible by 4, so the proposed converse is false.

A7. Disprove.

The forward direction "divisible by 4 -> even" is true, but the reverse "even -> divisible by 4" is false (counterexample 6). Therefore the biconditional is false.

A8.

MFA is necessary for privileged access. If A = access and M = MFA, the policy is A -> M. MFA alone need not be sufficient.

OPTIONAL

Appendix F - Optional enrichment

Three places this logic reappears later in mathematics and computing.

1. Modus tollens preview

p -> q
¬q
therefore ¬p

This inference rule is powered by the contrapositive relationship.

2. Proof by contraposition

To prove p -> q, it can be easier to prove ¬q -> ¬p instead.

This becomes a formal proof method later in the course.

3. AI / rule systems

Rules often have an if-then form.

Care is needed: a rule p -> q does not license q -> p unless the reverse rule is also known.

References and acknowledgments

Primary mathematical reference: Epp 4th edition; local lecture scope follows the COMP233 course materials.

Textbook

Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., Section 2.2: Conditional Statements.

Original course materials:	Prof. Mustafa Jarrar - Discrete Mathematics course materials
Updated course resources:	COMP233 GitHub repository

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